

tests/dynamic/suites/ concurrency_race_suite.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**
Authored for P10. Honest-SKIPs (topology_unsupported, §11.4.69)
until the live dynamic stack lands; parse-clean + authored today.

Companion (§11.4.18) to the in-source documentation block
at the top of the script.

Overview

§11.4.169 **concurrency / race** suite for the `external_acl` helper.
Under heavy concurrent load it asserts: (a) every parallel request
gets a **correct, consistent** routing decision (no torn Redis read,
no cross-request bleed), (b) **no deadlock/hang** (every request
returns within the timeout), (c) **determinism** across the burst
(§11.4.50 — identical inputs → identical outcomes). Per-request %
{http_code} is captured to its own file (the B3 bluff fix — never a
job exit status).

Prerequisites

- Source library `tests/dynamic/lib/analyzer_common.sh` +
committed `tests/lib/evidence.sh`.
- `curl`, `awk`, `sort`, POSIX `sh`, `date`.
- For a real run: `HELIX_DYNAMIC_STACK=1` + `HELIX_PROXY_URL`.
- Optional direct-helper probe: `HELIX_ACL_HELPER_CMD` — a direct
invocation of the Go `external_acl` binary that reads "`<Host>\n`"
on stdin and prints "`OK tag=...`" / "`ERR`".

Usage examples

```
# Today (no live stack): honest SKIP, exit 0:  
bash tests/dynamic/suites/concurrency_race_suite.sh
```

```
# P10 live run with the direct-helper hammer:  
HELIX_DYNAMIC_STACK=1 HELIX_PROXY_URL=http://127.0.0.1:53128 \  
hammer -c 1000000 -d 1000000 -s 1000000 -t 1000000 -u 1000000 -v 1000000 -w 1000000
```

```
RACE_CONC=20 HELIX_ACL_HELPER_CMD='./bin/external_acl' \  
bash tests/dynamic/suites/concurrency_race_suite.sh
```

Env: RACE_CONC (default 20), RACE_TARGET (default http://target-a.internal/), RACE_HOST (default target-a.internal).

Edge cases

- **Live stack absent** → topology_unsupported SKIP, exit 0.
- **A hang/deadlock** leaves an empty code file → counted as hung; any hung > 0 fails the GREEN guard.
- **Same Host yields more than one decision** (distinct_tags > 1) under the direct-helper hammer → a race → FAIL.

§11.4.115 RED_MODE polarity

RED_MODE	What it asserts	PASS means
0 (default)	GREEN guard	all 200, 0 hung, ≤1 decision per Host (consistent + live)
1	RED baseline	the pre-fix helper races/hangs/returns inconsistent tags → defect reproduced

A RED_MODE=1 run with no race/hang reproduced is a finding, not a pass.

Internal behaviour

- #!/usr/bin/env bash, POSIX-clean (sh -n + bash -n, §11.4.67).
- Fires RACE_CONC parallel proxied requests → req.N.code; tallies 200s and hung (empty/zero-byte code files).
- When HELIX_ACL_HELPER_CMD is set, also hammers the helper directly in parallel and reduces helper.*.out to sort -u distinct OK tag/ERR decisions; distinct_tags > 1 for one Host = a race.
- Counts + decisions written to concurrency.evidence; evidence dir qa-results/p9-harness/concurrency_race_<run-id>/.

Related

- tests/dynamic/lib/analyzer_common.sh — sourced base.

- tests/dynamic/suites/run_all_suites.sh — orchestrates this suite.
- Constitution §11.4.169 / §11.4.85 / §11.4.50 / §11.4.69 / §11.4.115 / §12.6; design §4/§13.

Last verified

2026-07-01 — run with no live stack: honest SKIP, exit 0; sh -n + bash -n parse-clean. Live concurrent acl-helper drive is exercised in **P10**.